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**Subscales of the Barratt Impulsiveness Scale Differentially Relate to the Big Five Factors of Personality**

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### Abstract

The place of impulsiveness in multidimensional personality frameworks is still unclear. In particular, no consensus has been reached yet with regard to the relation of impulsiveness to Neuroticism and Extraversion. We aim to contribute to a clearer understanding of these relationships by accounting for the multidimensional structure of impulsiveness. In three independent studies, we related the subscales of the Barratt Impulsiveness Scale (BIS) to the Big Five factors of personality. Study 1 investigated the associations between the BIS subscales and the Big Five factors as measured by the NEO-FFI in a student sample ( $N=113$ ). Selective positive correlations emerged between motor impulsiveness and Extraversion and between attentional impulsiveness and Neuroticism. This pattern of results was replicated in Study 2 ( $N=132$ ) using a 10-item short version of the Big Five Inventory. In Study 3, we analyzed BIS and NEO-FFI data obtained from a sample of patients with pathological buying ( $N=68$ ). In these patients, the relationship between motor impulsiveness and Extraversion was significantly weakened when compared to the non-clinical samples. At the same time, the relationship between attentional impulsiveness and Neuroticism was substantially stronger in the clinical sample. Our studies highlight the utility of the BIS subscales for clarifying the relationship between impulsiveness and the Big Five personality factors. We conclude that impulsiveness might occupy multiple places in multidimensional personality frameworks, which need to be specified to improve the interpretability of impulsiveness scales.

Keywords: impulsiveness; Extraversion; Neuroticism; Barratt Impulsiveness Scale; Five-factor model; pathological buying

### **Subscales of the Barratt Impulsiveness Scale Differentially Relate to the Big Five Factors of Personality**

The concept of impulsiveness is central to various contemporary personality theories (Cloninger, Svrakic, & Przybeck, 1993; Gray, 1982; Tellegen, 1985; Whiteside & Lynam, 2001; Zuckerman & Cloninger, 1996; Zuckerman, Kuhlman, Thornquist, & Kiers, 1991). However, the place of impulsiveness in multidimensional personality frameworks like the Five-Factor Model (Costa & McCrae, 1992) is unclear (Revelle, 1997; Sharma, Kohl, Morgan, & Clark, 2013). The Eysencks originally conceptualized impulsiveness as a facet of Extraversion (Eysenck & Eysenck, 1967; see also Revelle, 1997, for a review). In contrast, impulsiveness constitutes a subfactor of Neuroticism in the model of Costa & McCrae (1992), with other aspects of impulsiveness being reflected in the Conscientiousness construct.

The Barratt Impulsiveness Scale (BIS, Patton, Stanford & Barratt, 1995) was explicitly designed to be orthogonal to Neuroticism (Whiteside & Lynam, 2001). In support of this notion (and of the original Eysenckian idea), the BIS sum score has been found to be strongly related to Extraversion but not to Neuroticism in a study by Bachorowski et al. (1985). However, in the study by Helmers et al. (1997), the BIS was associated with Neuroticism but not with Extraversion, thus rather supporting the classification by Costa and McCrae (1992). This conceptual confusion has been attributed to the heterogeneity of the impulsiveness construct (Romero et al., 1994; Gerbing et al., 1987). Accounting for this heterogeneity by analyzing distinct components of impulsiveness might provide a clearer picture of the relationship between impulsiveness and basic personality factors (Romero et al., 1994). In their later work, Eysenck and Eysenck (1977) took a first step towards this goal by differentiating between impulsiveness

in the broad sense and impulsiveness in the narrow sense, the former being more closely related to Extraversion while the latter rather correlated with Neuroticism.

Similarly, the BIS has been developed based on a multifactorial understanding of impulsiveness (Barratt, 1985). The 30 items of the most current version of the BIS, the BIS-11 (Patton et al., 1995), relate to three intercorrelated factors. Attentional Impulsiveness (IA) refers to the inability to focus on the task at hand (“I don’t ‘pay attention.’”), motor impulsiveness (IM) involves acting without thinking (“I act on the spur of the moment.”), and non-planning impulsiveness (INP) reflects a lack of future orientation (“I plan for job security.”[reverse coded]).

As IM is considered to be “very similar to what the Eysencks had identified as impulsiveness narrow” (Patton et al., 1995, p. 769), one would expect a relationship between IM and Neuroticism. In contrast, in the influential study by Whiteside and Lynam (2001), IA emerged as the BIS subscale most closely linked to the Impulsiveness facet of the Big Five Neuroticism scale (i.e., both scales loaded on the same factor, negative urgency) (see also Sharma, Markon, & Clark, 2014). Fifteen years after Whiteside and Lynam (2001) the relationship between the BIS subscales and the Big Five factors of personality postulated by Costa & McCrae (1992) still remains to be clarified.

Here, we aim to contribute to the characterization of this relationship by presenting correlational data from three independent studies. Study 1 investigated whether the BIS subscales are differentially linked to Extraversion, Neuroticism, Conscientiousness, Agreeableness, and Openness as measured by the NEO-FFI. In Study 2, we analyzed whether the pattern of associations found in Study 2 replicates when a 10-item short version for the assessment of the Big Five factors is used instead. In contrast to the first two studies, Study 3 allowed examining the correlations of interest in a clinical sample with pathologically increased levels of

impulsiveness, particularly in treatment-seeking patients with pathological buying. This behavioral addiction is characterized by strong urges to buy resulting in repetitive, uncontrolled, excessive purchases of consumer goods that are then rarely or never used (e.g., Billieux et al., 2008; Müller, Mitchell, & de Zwaan, 2015). Previous research emphasized the prominent role of trait impulsivity in the development and maintenance of pathological buying behavior (e.g., Billieux et al., 2008; Paula et al., 2015; Rose & Segrist, 2014). In combination, these three studies shall shed new light on the complex relationship between impulsiveness and the Big Five factors by accounting for the multidimensional structure of impulsiveness.

### **General Methods**

In all three studies, participants completed an established German translation of the BIS-11 (Preuss et al., 2008). All items were answered on a 4-point scale (Rarely/Never, Occasionally, Often, Almost Always/Always). Higher scale scores indicate higher levels of impulsiveness. Subscale scores were calculated based on Patton et al. (1995).

Pearson correlation coefficients for the associations between BIS subscales and the Big Five factors were calculated using SPSS Version 23. The level of significance was set to  $\alpha = .05$ . For the evaluation of the associations of interest, the level of significance was set to  $\alpha = .003$  (= the nominal alpha level divided by 15 for the number of tests) to account for the number of possible correlations between the three BIS subscales and the Big Five factors. Note that our analyses are exploratory because neither the content of the BIS subscales nor the previously reported results in this area allowed us to specify a priori hypotheses as to which BIS subscale might be related to which Big Five factor.

## Study 1

### Methods

We conducted a secondary analysis of questionnaire data from  $N = 113$  students (101 female, mean age = 21.84 yrs,  $SD = 3.80$  yrs) that were collected as part of a larger study (see also Lange & Eggert, 2014; Lange & Eggert, 2015). Participants were compensated with partial course credit. All participants completed established German versions of the NEO-FFI (Borkenau & Ostendorf, 1993) and the BIS-11 (Preuss et al., 2008). In the tested sample, internal consistency was  $\alpha = .82$ ,  $\alpha = .52$ ,  $\alpha = .68$ , and  $\alpha = .66$  for the BIS sum score and the subscales IA, IM, and INP, respectively. With regard to the NEO-FFI scales, internal consistency was  $\alpha = .72$ ,  $\alpha = .85$ ,  $\alpha = .82$ ,  $\alpha = .76$ , and  $\alpha = .74$ , for Extraversion, Neuroticism, Conscientiousness, Agreeableness, and Openness, respectively. Sensitivity analysis (using G\*Power 3.1.9.2, Faul, Erdfelder, Lang & Buchner, 2007) revealed that, given the sample size and the corrected alpha level of  $\alpha = .003$ , we were able to detect medium-sized correlations of at least  $r = .28$  with sufficient statistical power ( $1-\beta = .80$ ).

### Results and Discussion

Summary statistics and Pearson correlation coefficients for the associations between BIS subscales and NEO-FFI scales are displayed in Table 1. The BIS subscales converged with regard to their correlations with the Conscientiousness scale of the NEO-FFI: IA, IM, and INP were negatively associated with Conscientiousness. However, the BIS subscales substantially differed with regard to their associations with Extraversion and Neuroticism: IA was positively related to Neuroticism ( $r = .30$ ,  $p = .001$ ) but not to Extraversion ( $r = .03$ ,  $p = .758$ ), whereas IM was positively related to Extraversion ( $r = .29$ ,  $p = .002$ ) but not to Neuroticism ( $r = -.08$ ,  $p = .532$ ). These findings are limited by the relatively poor reliability of the IA scale. In order to arrive at

more robust and conclusive results, we examined whether the observed correlations could be replicated in Study 2.

[please insert Table 1 about here]

### Study 2

#### Methods

$N = 132$  students (118 females, mean age = 21.62 yrs,  $SD = 2.98$  yrs) participated for course credit. Data from a subset of those participants ( $n = 68$ ) were collected as part of a larger study (see also Lange & Eggert, 2014; Lange, Seer, Rapior, Rose, & Eggert, 2014). Participants completed the German version of the BIS as well as the German version of the Big Five Inventory-10 (BFI-10), a 10-item short version of the Big Five Inventory (Rammstedt & John, 2007). Within the BFI-10, each of the five personality factors is assessed with two items which had been selected to “represent the core (i.e., most prototypical) traits that define each Big Five domain” (Rammstedt & John, 2007, p. 205). In the tested sample, internal consistency was  $\alpha = .85$ ,  $\alpha = .65$ ,  $\alpha = .70$ , and  $\alpha = .73$  for the BIS sum score and the subscales IA, IM, and INP, respectively. With regard to the NEO-FFI scales, internal consistency was  $\alpha = .79$ ,  $\alpha = .68$ ,  $\alpha = .52$ ,  $\alpha = .41$ , and  $\alpha = .60$ , for Extraversion, Neuroticism, Conscientiousness, Agreeableness, and Openness, respectively. Sensitivity analysis (using G\*Power 3.1.9.2, Faul et al., 2007) revealed that, given the sample size and the corrected alpha level of  $\alpha = .003$ , we were able to detect medium-sized correlations of at least  $r = .26$  with sufficient statistical power ( $1 - \beta = .80$ ).

#### Results and Discussion

Summary statistics and Pearson correlation coefficients for the associations between BIS subscales and BFI-10 scales are displayed in Table 2. The results closely mirrored the pattern found in Study 1. Again, all three BIS subscales were negatively correlated with

Conscientiousness and again, IA and IM could be dissociated with regard to their relationship to Extraversion and Neuroticism. IA was positively related to Neuroticism ( $r = .35, p < .001$ ) but not to Extraversion ( $r = .08, p = .338$ ), whereas IM was positively related to Extraversion ( $r = .38, p < .001$ ) but not to Neuroticism ( $r = .04, p = .676$ ). Hence, the observed associations between the BIS subscales, Extraversion, Neuroticism, and Conscientiousness appear to be robust across studies and unaffected by the choice of measures for assessing the Big Five factors.

In a next step, in order to obtain reliable estimates of the size of these associations, we pooled the data across Study 1 and Study 2 yielding an overall sample size of  $N = 245$ . As demonstrated by Schönbrodt & Perugini (2013), sample sizes as large as  $N \geq 238$  are typically required for stable estimates of correlation coefficients in the field of personality and social psychology. Meeting this criterion, our cross-study analysis revealed medium-to-large effect sizes for the associations between Conscientiousness and IA ( $r = -.38$ ), IM ( $r = -.35$ ), and INP ( $r = -.38$ ). Similarly, medium effect sizes were found for the correlation between Neuroticism and IA ( $r = .33$ ) and for the correlation between Extraversion and IM ( $r = .33$ ).

[please insert Table 2 about here]

## Study 3

### Methods

To investigate the associations between BIS subscales and the Big Five factors in a clinical sample with increased impulsiveness, we conducted a secondary analysis of data that we had collected within an earlier study (Mueller et al., 2010). The sample consisted of 68 patients (58 females) seeking psychotherapy treatment for pathological buying, aged between 19 and 61 years ( $M = 40.57$  yrs,  $SD = 10.61$  yrs). In the tested sample, internal consistency was  $\alpha = .83$ ,  $\alpha = .65$ ,  $\alpha = .73$ , and  $\alpha = .66$  for the BIS sum score and the subscales IA, IM, and INP, respectively.

With regard to the NEO-FFI scales, internal consistency was  $\alpha = .64$ ,  $\alpha = .82$ ,  $\alpha = .80$ ,  $\alpha = .70$ , and  $\alpha = .63$ , for Extraversion, Neuroticism, Conscientiousness, Agreeableness, and Openness, respectively. Sensitivity analysis (using G\*Power 3.1.9.2, Faul et al., 2007) revealed that, given the sample size and the corrected alpha level of  $\alpha = .003$ , we were able to detect medium-sized correlations of at least  $r = .35$  with sufficient statistical power ( $1-\beta = .80$ ).

## Results and Discussion

Summary statistics and Pearson correlation coefficients for the associations between BIS subscales and NEO-FFI scales are displayed in Table 3. Visual inspection revealed that all BIS subscales were again negatively related to Conscientiousness although the association between IM and Conscientiousness did not reach significance ( $r = -.25$ ,  $p = .037$ ). In addition, the positive correlation between IA and Neuroticism could be replicated ( $r = .57$ ,  $p < .001$ ). This correlation was significantly stronger than the  $r = .33$  correlation between IA and Neuroticism in the student samples ( $z = 2.18$ ,  $p = .029$ ) as indicated by a comparison of correlation coefficients based on Fisher r-to-z transformation (<http://vassarstats.net/rdiff.html>). In contrast, the correlation between IM and Extraversion was not significant ( $r = .07$ ,  $p = .597$ ) and significantly weaker than the  $r = .33$  correlation between IM and Extraversion in the student samples ( $z = 2.00$ ,  $p = .046$ , see also Figure 1).

[please insert Table 3 & Figure 1 about here]

## General Discussion

Using data from student and clinical samples, our analysis aimed at elucidating the place of impulsiveness in a multidimensional model of personality. Across three studies, the three subscales of the BIS could consistently be inversely linked to the Big Five factor Conscientiousness (cf. Vaidya, Latzman, Markon, & Watson, 2010). More surprisingly, the BIS

subscales IM and IA appeared to be differentially related to Extraversion and Neuroticism, suggesting that these facets of impulsiveness take up different places in a multidimensional personality framework. In non-clinical samples, the positive correlations between IM and Extraversion and between IA and Neuroticism emerged to be replicable, unaffected by the choice of Big Five measure, and medium in effect size. In a clinical sample involving individuals with elevated levels of impulsiveness, the relationship between IM and Extraversion was found to be weakened whereas the relationship between IA and Neuroticism was substantially stronger when compared to the non-clinical sample. In sum, these results may partially account for the discrepancies and inconsistencies in previous research on the relationship between impulsiveness and the Big Five personality factors.

Previous attempts to place impulsiveness in multidimensional personality frameworks did not arrive at a consensus as to whether impulsiveness was more closely related to Extraversion or to Neuroticism (Bachorowski et al., 1985; Eysenck & Eysenck, 1967; Helmers et al. 1997; Revelle, 1997). In accordance with Romero et al. (1994) our data suggest that this debate can be partially resolved by accounting for the complexity of the impulsiveness construct. It appears that the subscales of the BIS allow for a meaningful and fruitful distinction of different facets of impulsiveness. These subscales have already been shown to be differentially related to psychiatric disorders (e.g., Meule, 2013; Swann et al., 2008) and to behavioral measures of impulsiveness (Lange & Eggert, 2015; Enticott et al., 2006; Gorlyn et al., 2005; Keilp et al., 2005; Swann et al., 2002). The present set of studies further suggests that the BIS subscales can be dissociated with regard to their associations to Extraversion and Neuroticism. The robust correlation between IM and Extraversion might indicate that this BIS subscale captures the aspects of impulsiveness the Eysencks originally thought of when conceptualizing impulsiveness as a facet of Extraversion (maybe represented by items like “I am happy-go-lucky” [item 4]). In

contrast, the IA subscale might rather reflect those aspects of impulsiveness overlapping with the Neuroticism construct in Costa & McCrae's (1992) personality model (maybe represented by items like "I have racing thoughts" [item 6]).

While the selective associations between IM and Extraversion and between IA and Neuroticism could be observed irrespective of whether the NEO-FFI or the shorter BFI-10 were used for assessing the Big Five factors, the strength of these associations significantly varied between samples. The strengthened association between IA and Neuroticism in patients with compulsive buying is consistent with the results presented by Preuss et al. (2008) who found stronger correlations between impulsiveness and Neuroticism in alcohol-dependent and suicidal patients. One possible explanation for more pronounced relationships between impulsiveness and Neuroticism in clinical populations might be that some psychiatric conditions go along with increases in both impulsiveness and Neuroticism. As both these increments are related to the progression or severity of the disorder, they should be more closely related to each other than the variances of impulsiveness and Neuroticism in the healthy population. However, the sample differences across our three studies in the average levels of impulsiveness and Neuroticism (see Tables 1-3) were not fully compatible with this explanation. Whereas impulsiveness on the BIS indeed appeared to be more than a standard deviation higher in patients with pathological buying than in college students, the average level of Neuroticism was similar across studies. We can only speculate that, in the absence of quantitative differences, Neuroticism in impulsive patients might be of a different quality than in students. With regard to the weakened association between IM and Extraversion, inspection of the descriptive data across studies reveals that Extraversion was substantially reduced in the clinical sample. This reduction might have also affected the facets of Extraversion related to impulsiveness and, as a consequence, a decrease of the positive

correlation between impulsiveness and Extraversion could be observed in individuals with pathological buying.

It is important to note that differences in correlation coefficients between our non-clinical and clinical sample cannot unambiguously be attributed to disease-related factors. For example, participants in Study 3 were also considerably older than the participants included in the first two studies. As age-related changes have been demonstrated with regard to both the Big Five factors (Soto, John, Gosling, & Potter, 2011) and impulsiveness (Spinella, 2007), age differences between our studies cannot be excluded as a possible source of between-study heterogeneity. Future studies involving carefully matched control groups are required to establish how sample characteristics modulate the relationship between impulsiveness and the Big Five factors. In a similar vein, these studies should also investigate whether the reported correlational pattern can be generalized beyond the rather specific populations (i.e., college students and patients with pathological buying) examined in our studies. The proportion of male participants was rather low within all our samples and we cannot rule out the possibility of gender differences in the associations between facets of impulsiveness and the Big Five factors. However, previous studies have suggested that the correlations of impulsiveness with Extraversion and Neuroticism do not differ substantially between male and female participants (Eysenck & Eysenck, 1977, 1978).

In sum, our three studies have demonstrated the utility of the BIS subscales for clarifying the relationship between impulsiveness and the Big Five personality factors. We conclude that impulsiveness might, in fact, occupy multiple places in multidimensional personality frameworks which need to be specified to improve our understanding of impulsiveness and the interpretability of impulsiveness scales.

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ACCEPTED MANUSCRIPT

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# IMPULSIVENESS AND THE BIG FIVE

Table 1.

*Summary statistics and Pearson correlation coefficients for associations between the five factors of personality as measured by NEO-FFI and impulsiveness as measured by the BIS and its subscales in Study 1 (N = 113)*

| Measure                          | M    | SD   | NEO-FFI scales |             |              |             |             | BIS scales |      |      |   |
|----------------------------------|------|------|----------------|-------------|--------------|-------------|-------------|------------|------|------|---|
|                                  |      |      | 1              | 2           | 3            | 4           | 5           | 6          | 7    | 8    | 9 |
| 1 NEO-FFI_extraversion           | 3.57 | 0.46 | -              |             |              |             |             |            |      |      |   |
| 2 NEO-FFI_neuroticism            | 2.69 | 0.64 | -.43*          | -           |              |             |             |            |      |      |   |
| 3 NEO-FFI_conscientiousness      | 3.75 | 0.51 | .13            | -.27        | -            |             |             |            |      |      |   |
| 4 NEO-FFI_agreeableness          | 3.88 | 0.49 | .20            | -.03        | -.01         | -           |             |            |      |      |   |
| 5 NEO-FFI_openness               | 3.73 | 0.51 | -.07           | -.01        | -.17         | .00         | -           |            |      |      |   |
| 6 BIS-total                      | 2.01 | 0.30 | .24            | .06         | -.50*        | -.01        | -.03        | -          |      |      |   |
| 7 BIS-attentional impulsiveness  | 1.97 | 0.36 | <b>.03</b>     | <b>.30*</b> | <b>-.43*</b> | <b>-.06</b> | <b>.04</b>  | .72*       | -    |      |   |
| 8 BIS-motor impulsiveness        | 1.99 | 0.36 | <b>.29*</b>    | <b>-.08</b> | <b>-.36*</b> | <b>-.06</b> | <b>.04</b>  | .87*       | .51* | -    |   |
| 9 BIS-non-planning impulsiveness | 2.06 | 0.37 | <b>.23</b>     | <b>.01</b>  | <b>-.45*</b> | <b>.09</b>  | <b>-.13</b> | .84*       | .39* | .59* | - |

*Note:* Bold font highlights the 15 correlation coefficients of interest for the present analysis. \*Correlation is significant at the corrected two-tailed significance level of  $\alpha = .003$  ( $\alpha = .05$  divided by the number of correlation coefficients of interest).

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Table 2.

Summary statistics and Pearson correlation coefficients for the associations between BFI-10 scales and BIS subscales in Study 2 (N = 132)

| Measure                          | M    | SD   | BFI-10 scales |             |              |             |             | BIS scales |      |      |   |
|----------------------------------|------|------|---------------|-------------|--------------|-------------|-------------|------------|------|------|---|
|                                  |      |      | 1             | 2           | 3            | 4           | 5           | 6          | 7    | 8    | 9 |
| 1 BFI-10_extraversion            | 3.54 | 0.95 | -             |             |              |             |             |            |      |      |   |
| 2 BFI-10_neuroticism             | 2.93 | 0.97 | -.22          | -           |              |             |             |            |      |      |   |
| 3 BFI-10_conscientiousness       | 3.73 | 0.77 | -.03          | -.06        | -            |             |             |            |      |      |   |
| 4 BFI-10_agreeableness           | 3.36 | 0.77 | .13           | -.02        | -.11         | -           |             |            |      |      |   |
| 5 BFI-10_openness                | 3.78 | 0.99 | -.01          | .04         | -.14         | -.17        | -           |            |      |      |   |
| 6 BIS-total                      | 1.97 | 0.32 | .29*          | .21         | -.43*        | .15         | .06         | -          |      |      |   |
| 7 BIS-attentional impulsiveness  | 1.96 | 0.40 | <b>.08</b>    | <b>.35*</b> | <b>-.36*</b> | <b>-.06</b> | <b>-.04</b> | .73*       | -    |      |   |
| 8 BIS-motor impulsiveness        | 1.93 | 0.35 | <b>.38*</b>   | <b>.04</b>  | <b>-.34*</b> | <b>.22</b>  | <b>.13</b>  | .88*       | .49* | -    |   |
| 9 BIS-non-planning impulsiveness | 2.02 | 0.40 | <b>.24</b>    | <b>.17</b>  | <b>-.37*</b> | <b>.19</b>  | <b>.05</b>  | .88*       | .43* | .68* | - |

Note: Bold font highlights the 15 correlation coefficients of interest for the present analysis. \*Correlation is significant at the corrected two-tailed significance level of  $\alpha = .003$  ( $\alpha = .05$  divided by the number of correlation coefficients of interest).

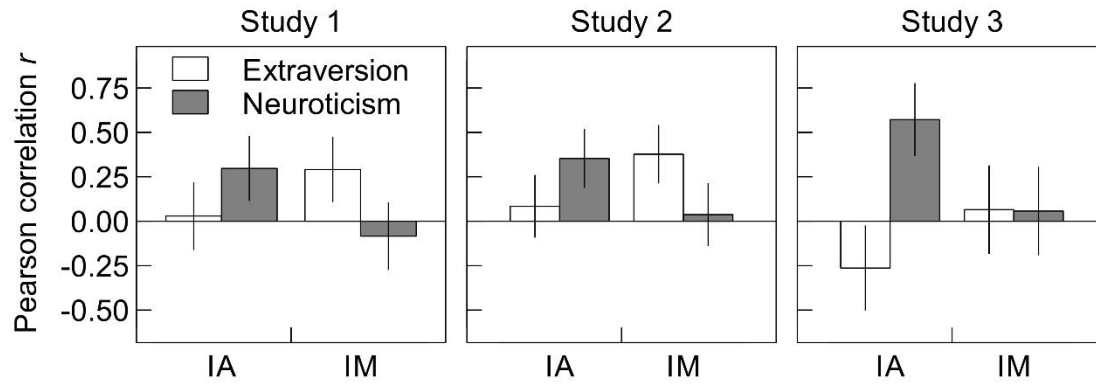
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Table 3.

*Summary statistics and Pearson correlation coefficients for the associations between NEO-FFI scales and BIS subscales in Study 3 (N = 68)*

| Measure                          | M    | SD   | NEO-FFI scales |             |              |             |             | BIS scales |      |      |   |
|----------------------------------|------|------|----------------|-------------|--------------|-------------|-------------|------------|------|------|---|
|                                  |      |      | 1              | 2           | 3            | 4           | 5           | 6          | 7    | 8    | 9 |
| 1 NEO-FFI_extraversion           | 2.02 | 0.90 | -              |             |              |             |             |            |      |      |   |
| 2 NEO-FFI_neuroticism            | 2.71 | 0.68 | -.52*          | -           |              |             |             |            |      |      |   |
| 3 NEO-FFI_conscientiousness      | 2.26 | 0.65 | .44*           | -.22        | -            |             |             |            |      |      |   |
| 4 NEO-FFI_agreeableness          | 2.34 | 0.52 | .30            | -.21        | .25          | -           |             |            |      |      |   |
| 5 NEO-FFI_openness               | 2.29 | 0.50 | .10            | -.26        | .15          | .01         | -           |            |      |      |   |
| 6 BIS-total                      | 2.54 | 0.37 | -.13           | .28         | -.54*        | -.28        | -.25        | -          |      |      |   |
| 7 BIS-attentional impulsiveness  | 2.47 | 0.50 | <b>-.26</b>    | <b>.57*</b> | <b>-.39*</b> | <b>-.20</b> | <b>-.26</b> | .72*       | -    |      |   |
| 8 BIS-motor impulsiveness        | 2.53 | 0.46 | <b>.07</b>     | <b>.06</b>  | <b>-.25</b>  | <b>-.22</b> | <b>-.11</b> | .85*       | .45* | -    |   |
| 9 BIS-non-planning impulsiveness | 2.61 | 0.45 | <b>-.16</b>    | <b>.12</b>  | <b>-.65*</b> | <b>-.25</b> | <b>-.24</b> | .81*       | .36* | .54* | - |

*Note:* Bold font highlights the 15 correlation coefficients of interest for the present analysis. \*Correlation is significant at the corrected two-tailed significance level of  $\alpha = .003$  ( $\alpha = .05$  divided by the number of correlation coefficients of interest).



*Figure 1.* Pearson correlation coefficients for the associations between the Big Five factors Extraversion and Neuroticism on the one hand and the Barratt Impulsiveness Scale subscales attentional impulsiveness (IA) and motor impulsiveness (IM) on the other hand. Error bars indicate 95% confidence intervals.